

- (1) Find conjugacy classes and fusion classes of A_5 .
- (2) Determine conjugacy classes and fusion classes of $\mathbb{Z}_{p^e}^d, \mathbb{Z}_{p^a} \times \mathbb{Z}_{p^b}$.
- (3) Let $G = D_{2n}$, a dihedral group of order $2n$. Find $\text{Aut}(G)$, and determine the conjugacy classes and fusion classes of elements of G .
- (4) Let G be a p -group, where p be a prime.
 - (a) Prove that the center $\mathbf{Z}(G) \neq 1$.
 - (b) Classify groups of order p^3 .
- (5) Let $G = \langle a, b \mid |a| = 2^e, |b| = 2, a^b = a^{2^{e-1}-1} \rangle$, where $e \geq 4$. Prove that G contains a subgroup which is isomorphic to $\mathbb{Z}_{2^e}$, D_{2^e} and Q_{2^e} .
- (6) Let G be a finite group of even order. Show that G has an odd number of elements of order 2.
- (7) Prove that A_n with $n \geq 5$ is a simple group.
- (8) Let G be a cyclic group of order n . Determine $\text{Aut}(G)$.
- (9) Let $G = \mathbb{Z}_p^d$, where p is a prime and d is a positive integer. Then $\text{Aut}(G) \cong \text{GL}(d, p)$.
- (10) Let $G = T^d$, where T is a nonabelian simple group. Then $\text{Aut}(G) = \text{Aut}(T) \wr S_d$, and

$$\text{Hol}(G) = \text{Hol}(T^d) = (T \rtimes \text{Aut}(T)) \wr S_d = (\check{G} \times \hat{G}).(\text{Out}(T) \wr S_d).$$

In particular, if $G = T$, then $\text{Hol}(T) = T \rtimes \text{Aut}(T) = (\check{T} \times \hat{T}).\text{Out}(T)$.

- (11) Let G be a finite group, and $M \triangleleft G$. Prove that G acts transitively on $\text{Syl}_p(M)$ where p is a prime divisor of $|M|$, and $G = MN_G(M \cap P)$ for $P \in \text{Syl}_p(G)$.
- (12) Let H, K be subgroups of a group G . Prove that $G = HK$ if and only if $|H||K| = |G||H \cap K|$.
- (13) If $H, K < G$ such that $|G : H|$ and $|G : K|$ are coprime, prove $G = HK$.
- (14) Let $G = A_n \times A_n$, with $n \geq 5$ odd, be a primitive permutation group of HS type. Prove that G has a regular subgroup which is isomorphic to $\mathbb{Z}_n \times A_{n-1}$.
- (15) State and prove Sylow's theorem.
- (16) Show that a group of order $351 = 3^3 \cdot 13$ has a normal Sylow subgroup.
- (17) Find two different Sylow 2-subgroups of S_5 and find an element that conjugates one to the other.
- (18) Prove that a group of order $255 = 3 \cdot 5 \cdot 17$ is abelian.
- (19) Let p, q, r be distinct primes with $p > q > r$. Classify groups of order pqr .
- (20) Assume that $M \triangleleft G$. Show that $G/\mathbf{C}_G(M) \lesssim \text{Aut}(M)$.
- (21) Let G be a group of order 72, and assume that the center $\mathbf{Z}(G) = 1$. Prove that G is isomorphic to one of the following groups:

$$\mathbb{Z}_3^2:\mathbb{Z}_8, \mathbb{Z}_3^2:\mathbb{Q}_8, \mathbb{Z}_3^2:D_8, (A_4 \times \mathbb{Z}_3):\mathbb{Z}_2.$$

- (22) Let T be a simple group which is 2-transitive on Δ , and let $G = T \wr S_3$ acts on $\Omega = \Delta^3$ in product action. Prove that G is of rank 4.
- (23) Let $T = A_n$, and let $G = T:S_n = (T \times T):S_2$ be a primitive permutation group of type HS. Find $\text{rank}(G)$.
- (24) Let $T = A_5$, and $N = T^k$. Determine the orbits of $\text{Aut}(N)$ acting on the elements of N .

- (25) Let G be a primitive permutation group of rank 3. Prove that G is affine, almost simple, or $T \wr S_2$ acting on Δ^2 in product action with T being 2-transitive on Δ .
- (26) A transitive permutation group on Ω is called *bi-quasiprimitive* if each of its non-trivial normal subgroups has at most two orbits on Ω , and there is one normal subgroup which has two orbits.
- Give examples of bi-quasiprimitive permutation groups.
 - Prove that, a transitive permutation group is bi-quasiprimitive if and only if each of its minimal normal subgroups has exactly two orbits.
 - How many minimal normal subgroups does a bi-quasiprimitive group have?
- (27) For a transitive permutation group G on Ω and $\omega \in \Omega$, an orbit of the stabilizer G_ω on $\Omega \setminus \{\omega\}$ is called a *suborbit* of G .
- Let G be a primitive permutation group which has a suborbit of length 2. Prove that G is a dihedral group of order $2p$ with p prime.
 - Let $G \leq \text{Sym}(\Omega)$ be primitive which has a suborbit of length 3 or 4. Prove that the stabilizer G_ω is a $\{2, 3\}$ -group.
 - Let $G \leq \text{Sym}(\Omega)$ be a p -group with p prime. Show that each suborbit of G has length being a p -power.
- (28) A group is called *generalised Hamiltonian* if each subgroup is not core-free. For example, a group $A \times Q_{2^n}$ where A is abelian and Q_{2^n} is generalised quaternion is a generalised Hamiltonian group.
- Prove that an abelian transitive permutation group is regular.
 - Show that a transitive permutation Hamiltonian group is regular.
 - Let G be a permutation group which contains a transitive dihedral subgroup. Prove that G contains a cyclic regular subgroup or a dihedral regular subgroup.
- (29) Let G be a finite solvable group whose Sylow subgroups are cyclic or dihedral. Describe the structure of G .
- (30) Let G be a finite solvable group. If $F(G)$ is a p -group, then $F(G/F(G))$ is a p -group.
- (31) Let G be solvable and $\Phi(G) = 1$. Prove the following statements.
- $F(G)$ is the only maximal abelian normal subgroup of G .
 - If G has a unique minimal normal subgroup N , then $F(G) = N$.
- (32) Let G be a group of order p^3 . Determine G' , $\mathbf{Z}(G)$, and $\Phi(G)$.
- (33) Let $A = \langle a \rangle = \mathbb{Z}_4$ and $B = Q_8$ with $\mathbf{Z}(B) = \langle c \rangle$. Let $G = A \circ B$ be the central product of A and B , namely, $G = (A \times B)/\langle a^2c \rangle$. List all the involutions of G , namely, elements of G of order 2.
- (34) State and prove Hall's theorems for solvable groups.
- (35) Let G be a finite solvable group with $\pi(G) = \{p_1, \dots, p_t\}$. Let $G_{p'_i}$ be a Hall p'_i -subgroup, with $1 \leq i \leq t$, and let $\mathcal{K} = \{G_{p'_1}, \dots, G_{p'_t}\}$. Prove
- for any $I \subset \{1, \dots, t\}$, the intersection $\bigcap_{i \in I} G_{p'_i}$ is a Hall subgroup;
 - G has Sylow p_i -subgroups G_{p_i} such that $G_{p_i}G_{p_j} = G_{p_j}G_{p_i}$.